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Rung for Real, Almost Never

A Cross-Portfolio Taxonomy of Genuine Fire-Alarm Call-Point Triggers

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Background

Every break-glass call point in a building portfolio is tested on a fixed monthly schedule. Little is known about how often, once installed, one is ever triggered for the reason it exists — and what, in practice, actually does the triggering.

Aims

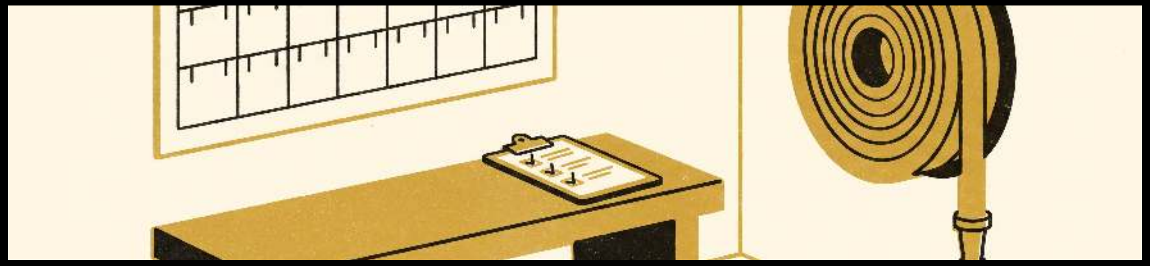
- Establish how many of a large portfolio's call points have ever logged a non-test activation
- Build a typology of the genuine triggers that do occur, against scheduled tests and false alarms
- Test whether call-point age predicts genuine-trigger likelihood

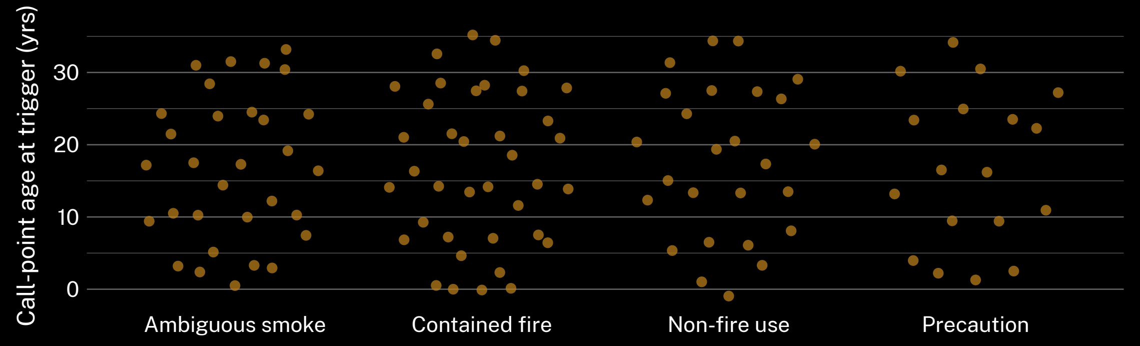
Methods

- 214 call points across 41 commercial and residential buildings; activation logs audited, 2020–2026
- Activations classified scheduled test / false alarm / genuine trigger by two independent coders ($\kappa = 0.86$)
- Genuine triggers typologised inductively; call-point age modelled as a predictor

Results

Of 3,142 logged activations, 96.4% are scheduled tests. The remaining 113 resolve into four recurring trigger types, with no association between call-point age and trigger type.





113 genuine activations logged across 41 buildings, 2020–2026, cluster into four trigger types; call-point age at time of trigger spans 0–34 years with no visible clustering by type.

Discussion & conclusions

A compliance record built almost entirely from its own scheduled tests reads as complete without describing readiness. The four-type typology clusters around ordinary hazards, not the fire scenario the mechanism was installed for. Next: audit whether portfolios with higher test-completion rates also report faster response once a genuine trigger occurs.

References

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Activation logs de-identified at building level; no occupant or reporting-party details retained. Supported by an Adaptive Metrics Lab seed allocation.

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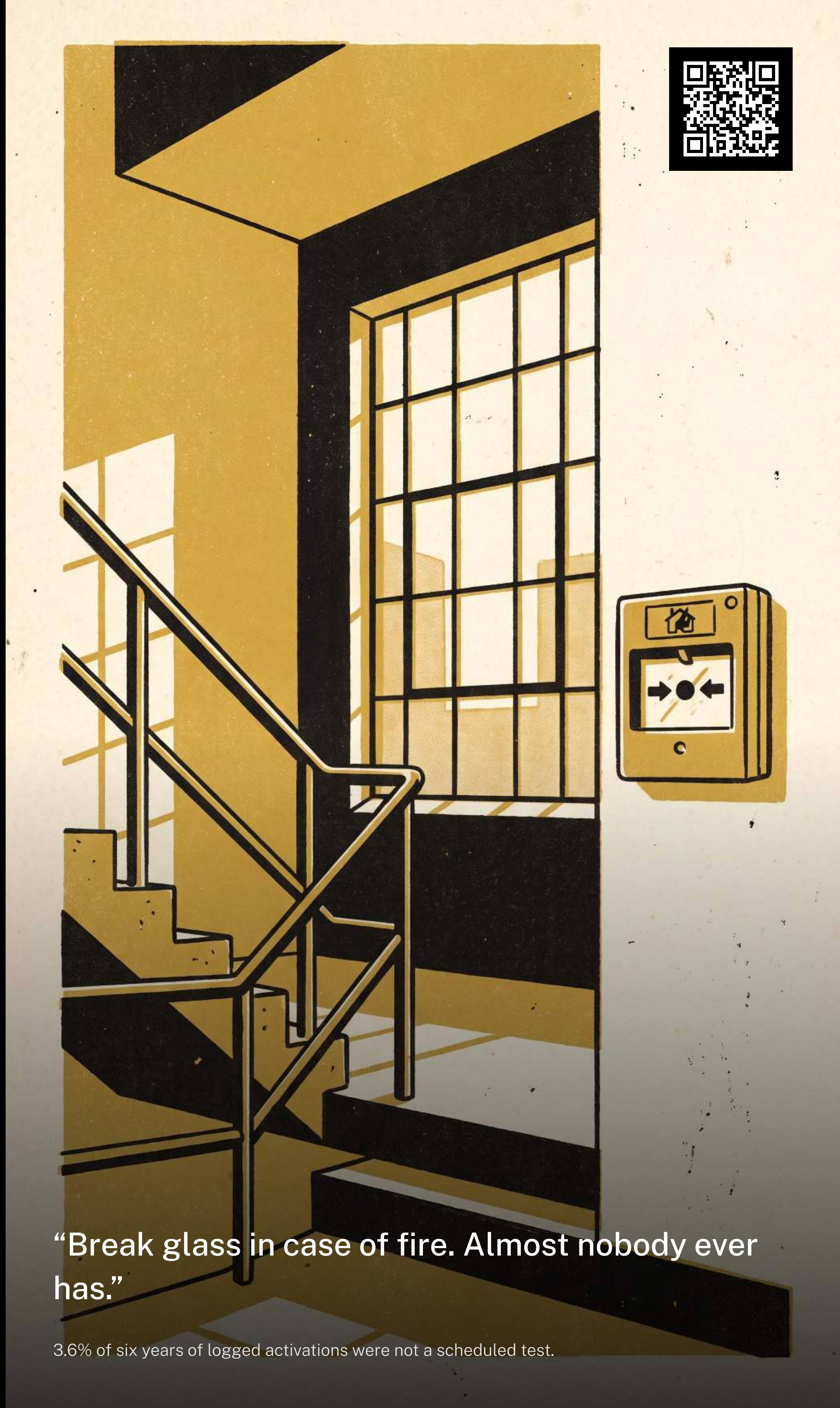

"Break glass in case of fire. Almost nobody ever has."

3.6% of six years of logged activations were not a scheduled test.